Lecture 5: Large Sample Properties

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STAT-S520 Introduction to Statistics

Outline

Random Samples

Large Sample Properties

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Random Samples

i.i.d. Random sample

- n independent random variables $X_1, \ldots, X_n$ are collected
- $X_1, \ldots, X_n$ follow the same distribution denoted by P (essentially, same CDF), so they are identically distributed.
- The shorthand notation i.i.d. means "independent and identically distributed"
- The notation is
$$X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} P,$$
- n is referred to as the sample size
- P is referred to as the population

Random Samples

Some Notation

- When we use alphabetical letters to denote variables, capital letters always denote random variables
- E.g., X, X_1, X_2, Y, Z, etc...
- In contrast, their lower case counterparts represent a realization or an observed value of the corresponding random variable, e.g., x, x_1, x_2, y, z, etc...
- Pure greek letters typically represent parameters that is associated with the population distribution, e.g., θ, μ, σ, ρ, β, etc...

Random Samples

The relation between probability and statistics

- Probability: Given a specific distribution P, learn the behavior of its outcomes $X_1, \ldots, X_n$
- Statistics: Given a random sample $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} P$, learn the behavior of the population P

Mathematically:

- Given a random sample $X_1, \ldots, X_n : \overset{\text{i.i.d.}}{\sim} P$ with an unknown population P
- We are interested in using $X_1, \ldots, X_n$ to "estimate" a certain attribute of the population P
- The attribute is typically denoted by a function ψ of P

Random Samples

Examples of attributes of population:

- Let X follow the population distribution P and cdf F.
- Mean & Variance: $\mu = \mathbb{E}(X) = \int x f_P(x) dx = \psi(P)$ and $\sigma^2 = \int (x - \mu)^2 f_P(x) dx = \psi(P)$
- Quantiles & Median: For any $\alpha \in (0, 1)$ the α quantile of P is defined by $\psi(P) = q_\alpha$ if $F(q_\alpha) = P(X \leq q_\alpha) = \alpha$;
- Interquartile range: The interquartile range is defined by $\mathrm{iqr}(X) = q_3 - q_1$,

Random Samples

- For now let us assume that we have zero knowledge of P
- We only have access to the random sample
- How can we use the random sample to "guess" the population attribute $\psi(P)$?
- In this lecture, let's focus on a special (but also very important) population attribute: the population expected value, or the population mean, $\mu = EX_1$.
- The process of finding a reasonable function of the random sample to "guess" the population mean μ is referred to as **estimation**
- The resulting function of the random sample, which itself is also a random variable, is referred to as an **estimator**

Random Samples

- Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} P$ with $EX_1 = \mu$
- Consider the random variable

$$\bar{X} = \frac{1}{n}(X_1 + \ldots + X_n) = \frac{1}{n}\sum_{i=1}^{n} X_i$$

- $\bar{X}$ is referred to as the **sample mean**
- In what follows we will argue that the sample mean is a reasonable estimate for the population expected value, i.e., $\mu = EX_1$
- $\bar{X}$ will be revisited multiple times in this course.

Outline

Random Samples

Large Sample Properties

Large Sample Properties

- Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} P$ be a random sample
- We now focus on the behavior of the sample mean $\bar{X}$ when the sample size n gets large
- So we denote $\bar{X} = \bar{X}_n$ to indicate the dependence on the sample size n

Large Sample Properties

- Consider the sample mean $\bar{X}_n$ as an estimator for the population mean $\mu = EX_1$
- By the linearity of the expected value,

$$E\bar{X}_n = E\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n}\sum_{i=1}^{n} EX_i = EX,$$

- This means that expectation of the sample mean is the same as the population mean

Large Sample Properties

- **More importantly,** by the independence of $X_1, \ldots, X_n$,

$$\begin{aligned}
\mathrm{var}(\bar{X}_n) &= \mathrm{var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) \\
&= \frac{1}{n^2}\mathrm{var}\left(\sum_{i=1}^{n} X_i\right) \\
&= \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{var}(X_i) = \frac{1}{n}\mathrm{var}(X),
\end{aligned}$$

- The above expression means that the variance of the sample mean decreases as the sample size n increases.
- This phenomenon is referred to as **averaging decrease variation**, because variance is a measurement of the variation of a random variable

Large Sample Properties

Convergence in probability:

- Let $(Y_n)_{n=1}^{\infty}$ be a sequence of random variables
- Assume that they are defined on the same probability space
- Let c be a constant
- We say that the sequence of random variables $(Y_n)_{n=1}^{\infty}$ converges to c in probability, denoted as $Y_n \xrightarrow{P} c$, if for any $\epsilon > 0$,

$$\lim_{n \to \infty} P(|Y_n - c| > \epsilon) = 0.$$

- Intuitively, convergence in probability says that the the sequence of random variables $(Y_n)_{n=1}^{\infty}$ concentrate in a small neighborhood around c with large probability.

Large Sample Properties

Weak Law of Large Numbers (WLLN)

- Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} P$ with $EX_1 = \mu < \infty$
- EX_1 is finite
- Assume that n can get arbitrarily large
- Let be the sample mean of $(X_1, \ldots, X_n)$

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i$$

- **Then $\bar{X}_n \xrightarrow{P} \mu$ as $n \to \infty$.** More specifically,

$$\lim_{n \to \infty} P(|\bar{X}_n - \mu| > \epsilon) = 0.$$

- Intuition of WLLN: eventually the sample mean will be extremely close to the population mean with high probability.

Large Sample Properties

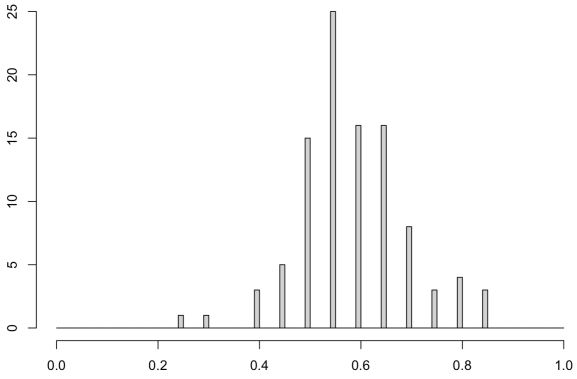
- WLLN says sample mean will be arbitrarily close to the population mean provided that the sample size is sufficiently large
- The implication of WLLN is that: If you use the sample mean $\bar{X}$ as a "guess", or formally, an estimator, for the population mean μ, then you are doing the right thing at the end of the day, because $\bar{X}$ will be arbitarily close to μ as n gets large

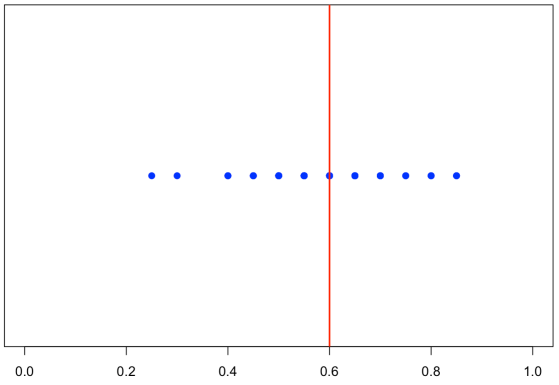
Large Sample Properties

Example

Suppose $X_i \sim \mathrm{Bernoulli}(p = 0.6)$ represents the correctness of the ith we ask GPT. We open a new chat with GPT to make sure that $X_1, \ldots, X_n$ are iid. We want to estimate the correct rate p, using $\bar{X}$.

1. Suppose we ask $n = 20$ questions. We show the locations and the histogram of 100 randomly generated $\bar{X}_{20}$.

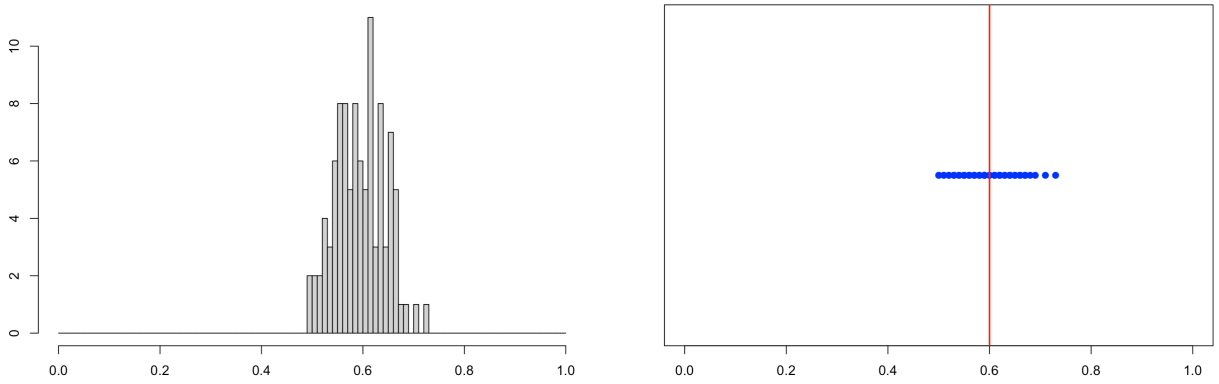




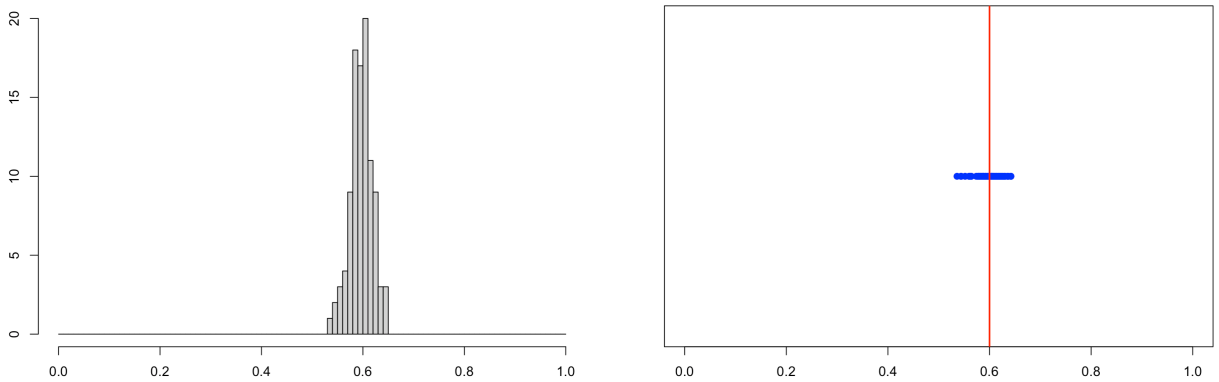
Large Sample Properties

Example

1. Suppose we ask $n = 50$ questions. We show the locations and the histogram of 100 randomly generated $\bar{X}_{50}$.



2. $\bar{X}_{500}$.



3. $P(|\bar{X}_n - \mu| > \epsilon)$ vanishes to 0 as n increases. In another word, $\bar{X}_n$ is more and more likely to get closer and closer to μ as n grows.

Large Sample Properties

- **Q:** Can we have more information about the property of $\bar{X}$ as a random variable other than that $\bar{X}_n$ converges to μ in stochastic senses?
- **A:** Yes!

Convergence in distribution to $\mathrm{N}(0,1)$:

- Let $(X_n)_{n=1}^{\infty}$ be a sequence of random variables
- They do not have to be defined on the same probability space, but we will use P to denote a generic probability measure that applies to all X_n's
- Let Φ denote the cdf of $\mathrm{N}(0,1)$
- We say that the sequence of random variables $(X_n)_{n=1}^{\infty}$ converges to $\mathrm{N}(0,1)$ in distribution, denoted as $X_n \xrightarrow{d} \mathrm{N}(0,1)$, if for any $x > \mathbb{R}$,

$$\lim_{n\to\infty} P(X_n \leq x) = \Phi(x) = F_{N(0,1)}(x).$$

- Intuition: For large n, the shape of the distribution of X_n is very close to $\mathrm{N}(0,1)$

Large Sample Properties

Central Limit Theorem (CLT)

- Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} P$ with $EX_1 = \mu$ and $\mathrm{var}(X_1) = \sigma^2$
- Assume that n can get arbitrarily large
- Let be the sample mean of $(X_1, \ldots, X_n)$

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i$$

- **Then as $n \to \infty$,**

$$\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \xrightarrow{\mathcal{L}} \mathrm{N}(0, 1)$$

- Intuition of CLT: The distribution of $\sqrt{n}(\bar{X}_n - \mu)/\sigma$ can be approximately regarded as $\mathrm{N}(0, 1)$ for large n
- **n cannot be too small!**

Large Sample Properties

Central Limit Theorem (CLT)

- Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} P$ with $EX_1 = \mu$ and $\mathrm{var}(X_1) = \sigma^2$
- **Then as $n \to \infty$,**

$$\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \xrightarrow{\mathcal{L}} \mathrm{N}(0, 1)$$

- Another way (or perhaps more useful way) to describe CLT is that

$$\text{distribution of } \bar{X}_n \approx \mathrm{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

- **n cannot be too small!**
- **Rule of Thumb:** Typically $n \geq 30$

Large Sample Properties

Example

Suppose that IUB faculty salaries follow an asymmetric distribution with population mean \$100,000 and sd \$55,000.

1. Suppose Yichi takes an i.i.d. sample of 100 salaries. Using the CLT, what's the probability the sample mean is between \$97,000 and \$103,000?
2. Suppose Yichi takes an i.i.d. sample of 1000 salaries. Using the CLT, what's the probability the sample mean is between \$97,000 and \$103,000?
3. Suppose Yichi takes an i.i.d. sample of 10 salaries. Without further information, is it possible to find the probability the sample mean is between \$97,000 and \$103,000?

Large Sample Properties

Example

Sampling model: $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} P$ with $EX_1 = 100,000$ and $\sigma^2 = \mathrm{var}(X_1) = 55000$

1. $n = 100$, which is large enough, so sample mean approximately follows $\mathrm{N}(\mu, \sigma^2/100)$:

```
pnorm(103000, 100000, 55000/sqrt(100)) -
  pnorm(97000, 100000, 55000/sqrt(100))
[1] 0.4145591
```

2. $n = 1000$, which is large enough, so sample mean approximately follows $\mathrm{N}(\mu, \sigma^2/1000)$:

```
pnorm(103000, 100000, 55000/sqrt(1000)) -
  pnorm(97000, 100000, 55000/sqrt(1000))
[1] 0.9154507
```

3. No. $n = 10$ is not a large sample size, particularly when the distribution is asymmetric.

Large Sample Properties

Example

- Suppose we take an i.i.d. sample of IUB faculty.
- But instead of measuring their exact salary, we just ask them: Do you earn at least $100,000?
- Let Y_i be a random variable that takes the value 1 if they answer yes, and takes the value 0 if they answer no.
- Each Y_i follows a Bernoulli distribution with success probability $p = P(X_1 \leq 100,000)$

Large Sample Properties

Example

- Let $S = Y_1 + \cdots + Y_n$, where $Y_1, \ldots, Y_n$ are i.i.d. Bernoulli random variables with parameter p.
- Well we already know what the distribution of S is! It's Binomial with parameters n and p!
- What the CLT tells us is that for (possibly very) large n, a Binomial random variable can be well-approximated by a Normal random variable with expected value $ES = np$ and variance $\mathrm{var}(S) = np(1-p)$.

Large Sample Properties

Example

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- Let $\bar{Y} = (Y_1 + \cdots + Y_n)/n$, the sample mean.
- For (possibly very) large n, the sample mean $\bar{Y}$ is approximately Normal, with $E\bar{Y} = p$ and $\mathrm{var}(\bar{Y}) = p(1-p)/n$
- In short,

$$\text{distribution of } \bar{Y} \approx \mathrm{N}\left(p, \frac{p(1-p)}{n}\right)$$

Large Sample Properties

Example

- Suppose 6.4% of faculty earn at least $200,000.
- Suppose we take an i.i.d. sample of 1000 faculty.
- Using the binomial, what's the probability that at least 50 out of the 1000 faculty earn at least $200,000?
- Using the normal distribution as an approximation, what's the probability that the sample proportion of faculty who earn at least $200,000 is at least 0.05?

Large Sample Properties

Example

- ```
  p = 0.064
  1 - pbinom(49, 1000, p)
  [1] 0.9730032
  ```
- ```
  1 - pnorm(0.05, p, sqrt(p*(1 - p)/1000))
  [1] 0.9647621
  ```
- With Bernoulli distributions with extreme values of p (close to 0 or 1), it can take a long time for the Central Limit Theorem to kick in.

Large Sample Properties

CLT with unknown variance

- In S520 it is more common to encounter the case where σ^2 is known
- Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} P$ with $EX_1 = \mu$ and $\mathrm{var}(X_1) = \sigma^2$
- **Q:** Is it possible to get information about the distribution of $\bar{X}_n$ without knowing σ^2?
- **A:** We can estimate σ^2 by the so-called sample variance (will be revisited in Chapter 9)

$$S_n^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2$$

- **Then as $n \to \infty$,**

$$\frac{\sqrt{n}(\bar{X} - \mu)}{S_n} \xrightarrow{\mathcal{L}} \mathrm{N}(0, 1)$$